Site and row model

Legalization snaps cells to a discrete grid

The idea

- Every cell sits on exactly one row with a lower-left coordinate aligned to site pitch one
- Width tells you how many consecutive sites the rectangle covers
- The golden packing is legal
- <!-- algorithm-walkthrough -->

Twelve-by-six chip, three rows

SITE & ROW MODEL

Twelve-by-six chip, three rows

Step 1 / 5 · empty-chip · module01-01-site-row-model

Explanation

Legalization lives on a discrete canvas: chip width twelve sites, height six (three rows of height two). Row bottoms sit at y equals zero, two, and four. Every movable cell must land on a row and align to site pitch one.

- CHIP_W = 12 · CHIP_H = 6
- Row bottoms: y = 0, 2, 4
- Lower-left origin for cell coordinates

```
chip: 12×6
rows: 3
rowH: 2
```

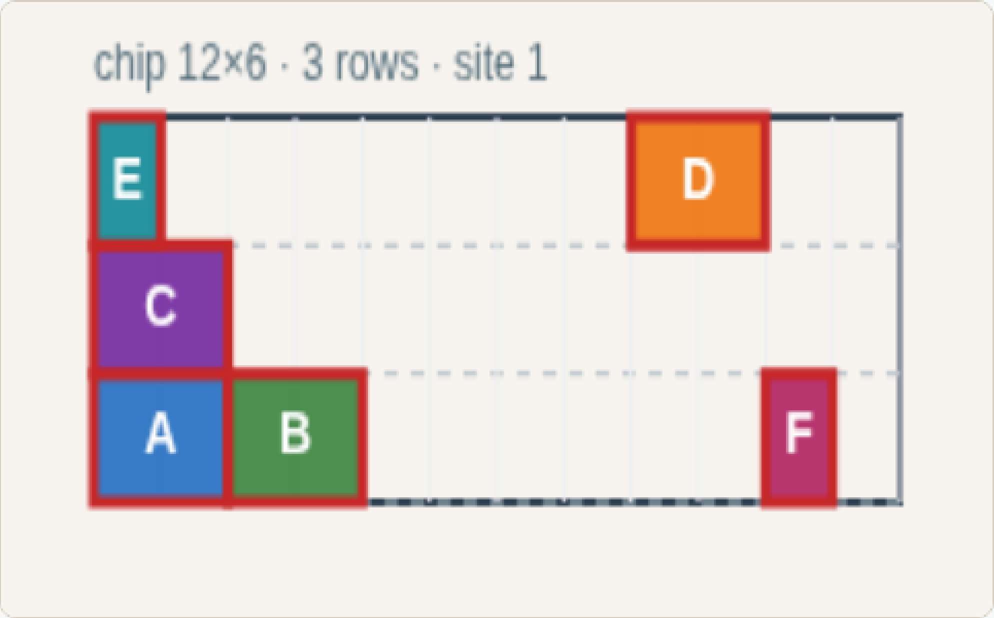
Twelve-by-six chip, three rows

Cell widths A–D = 2, E–F = 1

SITE & ROW MODEL

Cell widths A–D = 2, E–F = 1

Step 2 / 5 · cell-widths · module01-01-site-row-model



Explanation

Six cells A through F occupy ten sites total—two sites each for A–D and one each for E and F. Width drives how many consecutive sites a rectangle spans when you pack a row.

- Widths are integers in site units
- Total width 10 ≤ chip 12
- Same six-cell toy across all labs

```
A–D width: 2
E–F width: 1
total width: 10
```

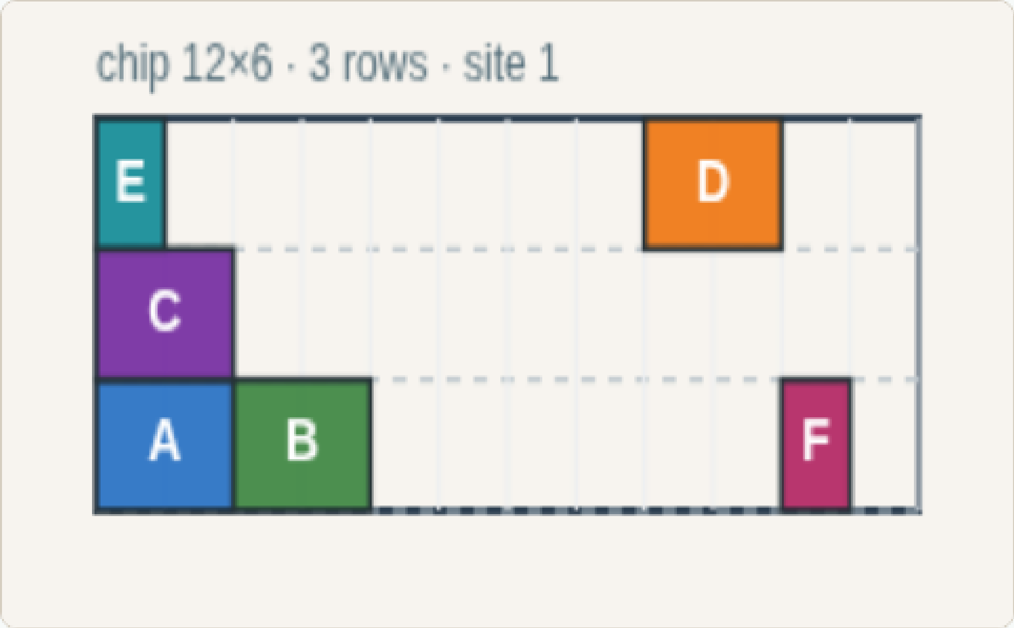
Cell widths A–D = 2, E–F = 1

Site pitch equals one

SITE & ROW MODEL

Site pitch equals one

Step 3 / 5 · site-pitch · module01-01-site-row-model



Explanation

Each vertical tick is one site. Legal x coordinates are integers; a width-two cell at x equals four covers sites four and five on its row. The golden packing shows legal site alignment.

- SITE_W = 1
- x must be integer-aligned
- Row lines dashed at y = 0, 2, 4

```
site pitch: 1
rows: 0, 2, 4
```

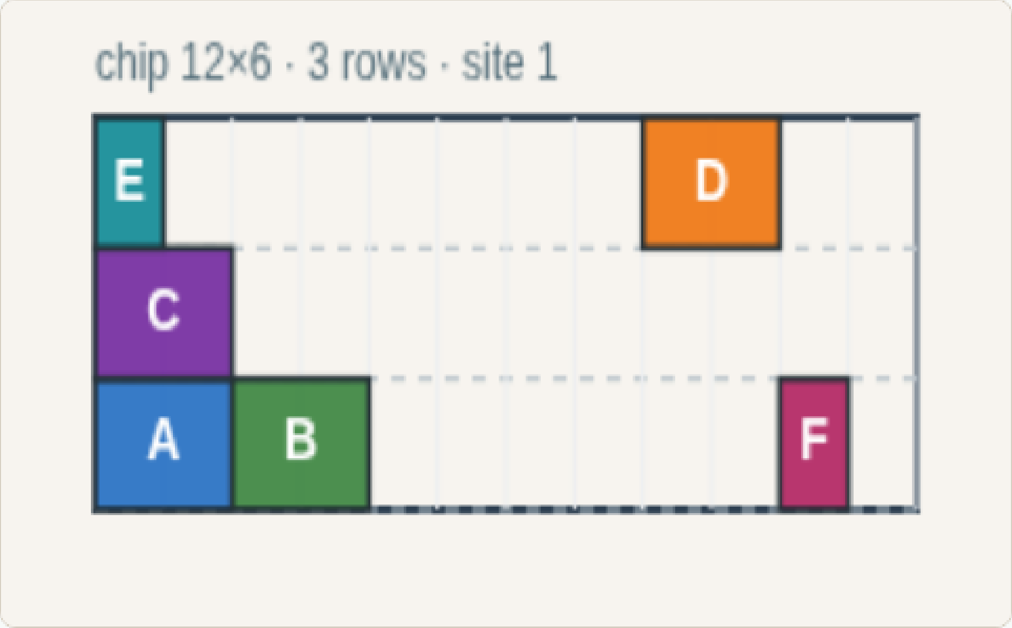
Site pitch equals one

Golden legal packing

SITE & ROW MODEL

Golden legal packing

Step 4 / 5 · golden-pack · module01-01-site-row-model



Explanation

The reference legal layout spreads A and B on row zero, C on row one, and E plus D on row top. No overlap, every cell on-row and site-aligned—this is the teaching golden.

- Row 0: A@0, B@2, F@10
- Row 2: C@0
- Row 4: E@0, D@8

```
legal: true
HPWL: 50
reason: ok
```

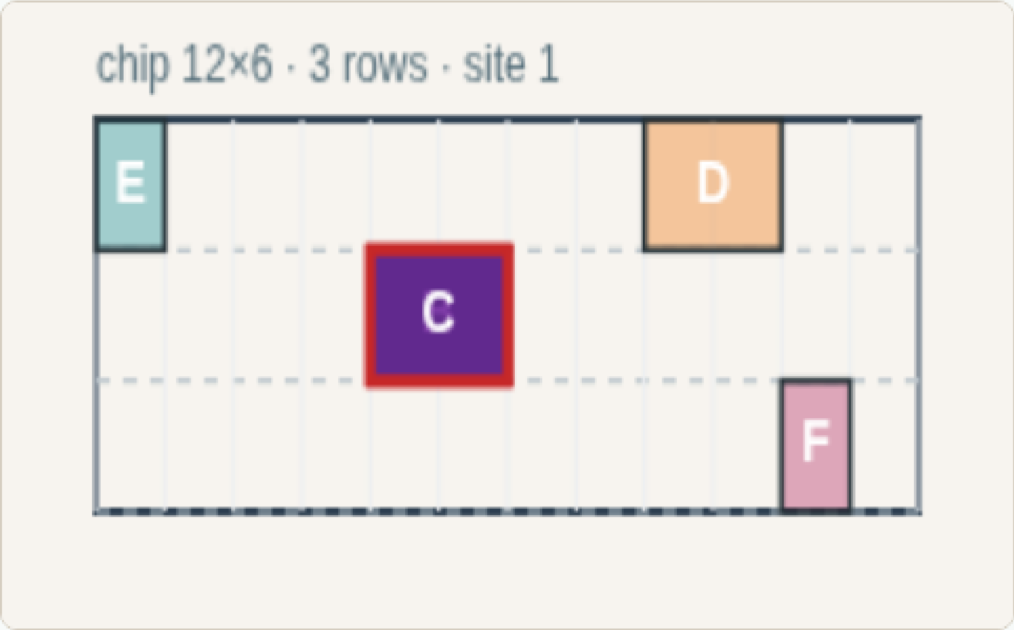
Golden legal packing

Contrast: illegal overlap stack

SITE & ROW MODEL

Contrast: illegal overlap stack

Step 5 / 5 · overlap-contrast · module01-01-site-row-model



Explanation

The overlap seed stacks A, B, and C at (4, 2) on the middle row—same width-two cells fighting for the same sites. Legality fails before you even discuss wirelength.

- A,B,C @ (4,2) stacked
- First failure: overlap A/B
- Legalization must repair this

```
legal: false
reason: overlap A/B
Next: legality metrics
```

Contrast: illegal overlap stack

Browser lab track

- In the browser lab track, open the **site-row-model** lab from the tools shelf
- Load the overlap or float starter, run the legalizer once
- Work the challenges that lock the goldens

Implement track

- In the implement track, open this module's examples and the course `common/` solvers
- Parse `tiny_legal.json`, run the algorithm with deterministic coordinates
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you're ready, take the short quiz, then continue to the next module